Supersymmetry is a more abstract symmetry principle that relates physical properties of elementary constituents that carry different amounts of spin. At best there are only hints from experimental results that the microworld incorporates this symmetry, but, for reasons discussed earlier, there is a strong belief that it does. It is certainly an integral part of string theory. In the 1990s, led by the pioneering work of Nathan Seiberg of the Institute for Advanced Study, physicists have realized that supersymmetry provides a sharp and incisive tool that can answer some very difficult and important questions by indirect means.

Even without understanding intricate details of a theory, the fact that it has supersymmetry built in allows us to place significant constraints on the properties it can have. Using a linguistic analogy, imagine that we are told that a sequence of letters has been written on a slip of paper, that the sequence has exactly three occurrences, say, of the letter "y," and that the paper has been hidden within a sealed envelope. If we are given no further information, then there is no way that we can guess the sequence—for all we know it might be a random assortment of letters with three y's like mvcfojziyxidqfqzyycdi or any one of the infinitely many other possibilities. But imagine that we are subsequently given two further clues: The hidden sequence of letters spells out an English word and it has the minimum number of letters consistent with the first clue of having three y's. From the infinite number of letter sequences at the outset, these clues reduce the possibilities to one word—to the shortest English word containing three y's: syzygy.

Supersymmetry supplies similar constraining clues for those theories that incorporate its symmetry principles. To get a feel for this, imagine that we are presented with a physics puzzle analogous to the linguistic puzzle just described. Hidden inside a box there is something —its identity is left unspecified—that has a certain force charge. The charge might be electric, magnetic, or any of the other generalizations, but to be concrete let's say it has three units of electric charge. Without further information, the identity of the contents cannot be determined. It might be three particles of charge 1, like positrons or protons; it might be four particles of charge 1 and one particle of charge -1 (like the electron), as this combination still has a net charge of three; it might be nine particles of charge one-third (like the up-quark) or it